

Properties of Logarithms

Applying the product, quotient, and power properties of logarithms to expand, condense, evaluate, and solve

Directions

- Use only the product, quotient, and power properties — no calculator except where a decimal answer is asked for.
- Every logarithm here has a positive argument, so the properties apply throughout.
- Name the property you use at each step; a correct answer with no visible property is only half the work.

1. Evaluate using the product property. $\log_2 8 + \log_2 4$

Answer: _____

2. Expand into separate terms, writing the constant term as a multiple of $\ln 2$. $\ln(8x)$

Answer: _____

3. Write as a single logarithm, then evaluate it. $2 \log_3 6 - \log_3 4$

Answer: _____

4. Evaluate using the product property. $\log_6 4 + \log_6 9$

Answer: _____

5. Expand into separate terms using the quotient property. $\log_2 \left(\frac{x}{32} \right)$

Answer: _____

6. Solve for x . $\log_3 x + \log_3 4 = \log_3 20$

Answer: _____

7. Write as a single logarithm. $3 \ln 2 + 2 \ln 5$

Answer: _____

8. (a) Expand $\log_2 \left(\frac{x^5}{16} \right)$ using the quotient and power properties.
(b) Use your expanded form to evaluate the expression when $x = 4$.

Answer: _____

9. Solve for x . $\log x - \log 3 = \log 7$

Answer: _____

10. A calculator gives $\log 2 \approx 0.3010$ and $\log 3 \approx 0.4771$. Using only the properties of logarithms and those two values, approximate $\log 48$ to three decimal places.

Answer: _____

11. (a) Write $3 \ln x - \ln y - 2 \ln z$ as a single logarithm.

(b) Evaluate your single logarithm when $x = 2$, $y = 4$, and $z = 1$. Leave the answer in exact form.

Answer: _____

12. Solve for x . $3 \log_2 x = \log_2 27$

Answer: _____